

$^{32}\text{S}(\alpha, \text{p})$ 2019Se09, 1973Go16

$J^\pi=0^+$ for ^{32}S ground state.

2019Se09: a 21-MeV α beam was produced from the 10-MV FN tandem Van de Graaff accelerator at Triangle Universities Nuclear Laboratory (TUNL). Targets were $15.9\text{-}\mu\text{g}/\text{cm}^2$ $^{\text{nat}}\text{S}$ in cadmium sulfide and $46.3\text{-}\mu\text{g}/\text{cm}^2$ $^{\text{nat}}\text{S}$ in antimony sulfide. Reaction products were momentum-analyzed by the TUNL high-resolution Enge split-pole magnetic spectrograph and detected using a high-resolution position-sensitive focal-plane detector consisting of two position-sensitive gas avalanche counters: a gas proportionality energy loss section and a residual energy scintillator (FWHM=24 keV). Measured $\sigma(E_p, \theta)$, at $\theta=10^\circ$ to 50° . Deduced levels, J , π from one-step finite range triton transfer FRESKO-DWBA analysis. Deduced the thermonuclear $^{34}\text{S}(\text{p}, \gamma)^{35}\text{Cl}$ reaction rate via Monte Carlo methods.

1973Go16: 19.991-, 20.025- and 20.284-MeV α beams were produced from the University of Notre Dame FN tandem Van de Graaff accelerator. Natural sulfur targets were prepared by vacuum evaporation of CdS onto $20\text{-}\mu\text{g}/\text{cm}^2$ carbon backings. Protons were analyzed with broad-range magnetic spectrographs at 15° , 30° , 70° , 90° , and 120° and detected using nuclear track plates at the focal planes. Measured $\sigma(E_p, \theta)$. Deduced 102 levels.

1971Au07: a 28.8-MeV α beam of 0.1-0.5 μA was produced from the Naval Research Laboratory (NRL) isochronous cyclotron. The target was made by evaporating natural PbS onto a $30\text{-}\mu\text{g}/\text{cm}^2$ carbon backing, and the effective ^{32}S thickness was $163\text{ }\mu\text{g}/\text{cm}^2$. Protons were detected using two counter telescopes with FWHM=250 keV. Measured $\sigma(E_p, \theta)$. Deduced levels, J , π , L-transfers, and J-dependence for the g.s. and 1763-keV level from local zero-range DWUCK-DWBA analysis of the measured $\sigma(\theta)$.

Other: **1996So21** (cross sections).

 ^{35}Cl Levels

<u>E(level)[†]</u>	<u>J^π &</u>	<u>L^a</u>	<u>Comments</u>
0		2	J^π : $3/2^+$ assumed in 1971Au07 for investigation of J-dependence of L=2 transfer.
1217.4 26			
1760.1 24		(2)	J^π : $5/2^+$ assumed in 1971Au07 for investigation of J-dependence of L=2 transfer.
			L: The parentheses are added by evaluators based on the DWBA fit in 1971Au07 .
2644.9 23			
2693?			
3001.5 31			
3158.2 25			
3945.2 23			E(level): weighted average of 3943.7 23 (1973Go16) and 3947.9 31 (2019Se09).
4056.9 27	($3/2^-$) ^b		E(level): 4059.12 quoted from 2011Ch48 and used for internal calibrations by 2019Se09 .
4111.1 29	($7/2^+$) ^b		E(level): weighted average of 4110.2 29 (1973Go16) and 4112.0 30 (2019Se09).
4177.9 24	($3/2^-$) ^b		E(level): weighted average of 4177.5 24 (1973Go16) and 4179 4 (2019Se09).
4346.6 24			E(level): 4347.82 quoted from 2011Ch48 and used for internal calibrations by 2019Se09 .
4625.3 28			E(level): weighted average of 4624.4 31 (1973Go16) and 4626.1 28 (2019Se09).
4770.4 26			E(level): weighted average of 4770.8 26 (1973Go16) and 4769.9 28 (2019Se09).
4859.2 [‡] 28			
4881.8 29			E(level): weighted average of 4883.1 29 (1973Go16) and 4878 5 (2019Se09).
5006.9 [‡] 27			
5158.9 29	($5/2^+$) ^b		E(level): weighted average of 5161.7 33 (1973Go16) and 5156.8 29 (2019Se09).
5209.5 28	($5/2^+$)		E(level): weighted average of 5206.6 37 (1973Go16) and 5211.1 28 (2019Se09).
5402.0 29			
5407.2	($11/2^-$) ^b		E(level): quoted from 2011Ch48 and used for internal calibrations by 2019Se09 .
5531 [‡] 5			
5586.0 26			E(level): 5576 tentative level from 1973Go16 .
5594.6 26			E(level): weighted average of 5591.8 32 (1973Go16) and 5596.5 26 (2019Se09).
5633.6 30			E(level): weighted average of 5633.1 32 (1973Go16) and 5634 3 (2019Se09).
5645 [‡] 4			
5653 [‡] 4			
5679.7 34			E(level): weighted average of 5677.7 34 (1973Go16) and 5684 5 (2019Se09).
5731.2 [‡] 25			
5757 [‡] 4			

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$^{32}\text{S}(\alpha, p)$ **2019Se09, 1973Go16** (continued) ^{35}Cl Levels (continued)

E(level) [†]	J ^π &	Comments
5808.1 31		E(level): weighted average of 5809.2 34 (1973Go16) and 5807.1 31 (2019Se09).
5829 [‡] 4	(5/2 ⁻) ^b	E(level): 5823 tentative level from 1973Go16.
5928 [‡] 3		E(level): other: 5927.4 35 (1973Go16).
6084.2 29	(13/2 ⁻) ^b	E(level): 6087.4 quoted from 2011Ch48 and used for internal calibrations by 2019Se09.
6104 [‡] 4		
6141 4		E(level): weighted average of 6140.2 40 (1973Go16) and 6142 4 (2019Se09).
6180 [‡] 3		
6224.4 30		E(level): weighted average of 6224.9 36 (1973Go16) and 6224 3 (2019Se09).
6283 [‡] 3	(5/2 ⁺)	
6379.3 25	(9/2 ⁻)	E(level): from 2019Se09. Other: 6379.0 34 (1973Go16).
6401.2 22	(1/2 ⁻)	E(level): weighted average of 6402.4 41 (1973Go16) and 6400.9 22 (2019Se09).
6429 [‡] 3	(1/2 ⁺)	E(level): 6427 tentative level from 1973Go16.
6475 [‡] 4		
6492 [‡] 3	(3/2 ⁺)	E(level): other: 6491.9 34 (1973Go16).
6549 [‡] 3	(1/2 ⁺)	
6659.1 30	(7/2 ⁺)	E(level): weighted average of 6656.0 31 (1973Go16) and 6662 3 (2019Se09).
6679.4 31	(1/2 ⁺)	E(level): weighted average of 6680.8 31 (1973Go16) and 6677 4 (2019Se09).
6781.3 30	(3/2 ⁻)	E(level): weighted average of 6782.8 32 (1973Go16) and 6780 3 (2019Se09).
6800 4		E(level): weighted average of 6802.1 42 (1973Go16) and 6795 6 (2019Se09).
6842 [‡] 4	(3/2 ⁺)	
6863 [‡] 3	(9/2 ⁺)	E(level): 6867 tentative level from 1973Go16.
6891.6 30	(9/2 ⁺)	E(level): weighted average of 6893.5 32 (1973Go16) and 6890 3 (2019Se09).
6948.6 34	(5/2 ⁺)	E(level): weighted average of 6947.5 34 (1973Go16) and 6950 4 (2019Se09).
6986.3 34	(9/2 ⁺)	E(level): weighted average of 6985.7 34 (1973Go16) and 6987 4 (2019Se09).
7066.2	(5/2 ⁺)	E(level): 7066.2 quoted from 2011Ch48 and used for internal calibrations by 2019Se09.
7105?		
7121.7 [#] 24	(5/2 ⁻)	E(level): weighted average of 7120.8 34 (1973Go16) and 7122.1 24 (2019Se09).
7180 [‡] 4	(7/2 ⁺)	E(level): other: 7179.0 43 (1973Go16).
7213.7 25		E(level): weighted average of 7210.1 39 (1973Go16) and 7215.2 25 (2019Se09).
7230.9 30	(3/2 ⁺)	E(level): weighted average of 7229.5 34 (1973Go16) and 7232 3 (2019Se09).
7301?		
7348 [‡] 3	(7/2)	E(level): other: 7348.3 44 (1973Go16).
7362 [‡] 3		
7413.8 30		E(level): weighted average of 7417.5 44 (1973Go16) and 7412 3 (2019Se09).
7447 [‡] 3		
7503 4		E(level): weighted average of 7503.0 37 (1973Go16) and 7502 5 (2019Se09).
7566.4 35		E(level): weighted average of 7567.6 35 (1973Go16) and 7564 5 (2019Se09).
7590 4		E(level): weighted average of 7587.2 40 (1973Go16) and 7595 6 (2019Se09).
7648.1 30		E(level): weighted average of 7650.1 40 (1973Go16) and 7647 3 (2019Se09).
7674.6 [‡] 25		E(level): 7671 tentative level from 1973Go16.
7709.9 [‡] 25		
7730?		
7744.1 [#] 30		E(level): weighted average of 7742.7 38 (1973Go16) and 7745 3 (2019Se09).
7770.3 33		E(level): weighted average of 7776.3 45 (1973Go16) and 7768.5 25 (2019Se09).
7799.8 30		E(level): weighted average of 7800.8 35 (1973Go16) and 7799 3 (2019Se09).
7868.5 35		E(level): from 1973Go16. Other: 7867 7 (2019Se09).
7901 4		E(level): weighted average of 7900.6 38 (1973Go16) and 7899 12 (2019Se09).
7978.6 35		
8000.7 41		
8019.2 35		
8075.1 46		

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$^{32}\text{S}(\alpha, \text{p})$ **2019Se09,1973Go16** (continued) ^{35}Cl Levels (continued)

E(level) [†]	E(level) [†]	E(level) [†]	E(level) [†]
8103.8 [#] 42	8654.1 41	9334.0 46	10117?
8121?	8698?	9376.0 41	10162.7 53
8171.7 42	8720.9 [#] 37	9475.5 43	10179.1 53
8210.2 46	8786.3 38	9508.1 [#] 41	10218.0 46
8246.0 42	8840.3 38	9525?	10395.2 54
8270.6 [#] 42	8887.5 41	9711.8 44	10443?
8292?	8957.5 44	9740.4 40	10463.2 [#] 49
8321.0 38	8996.9 40	9799.2 47	10517.1 54
8398.8 38	9027.4 38	9836.1 44	10548.3 54
8429.2 40	9085.7 38	9870?	10579.5 54
8482.3 38	9109.3 49	9968.6 [@] 52	10643.0 50
8505.1 47	9160.0 42	9992?	10717?
8532.4 43	9194.0 39	10009?	10732.1 [#] 50
8567.9 37	9264.9 39	10075.2 [@] 48	10759.1 55
8616.9 37	9316.2 39	10098?	

[†] From **1973Go16**, unless otherwise noted.

[‡] From **2019Se09**.

[#] Doublet in **1973Go16**.

[@] Triplet in **1973Go16**.

[&] From **2019Se09**, based on FRESKO-DWBA analysis of measured $\sigma(\theta)$, unless otherwise noted.

^a From **1971Au07**.

^b Original firm assignments in **2019Se09** are placed in parentheses by the evaluators when DWBA fits to the measured $\sigma(\theta)$ were poor or when only a few data points were available for fitting.